ExamGOAL
PYQ PDF

Free · With answer key

GATE ECE · NETWORK THEORY · CHAPTER-WISE PYQS

Network Theorems

GATE ECE Network Theory · Previous Year Questions with Answer Key

56	37	1988 – 2026	English
QUESTIONS	PAPERS COVERED	YEARS	LANGUAGE

INSIDE THIS PDF

Marks 1	25 questions
Marks 2	26 questions
Marks 5	5 questions
Answer key	at the end

Practise online

Every question with its full solution, free on **questions.examside.com**

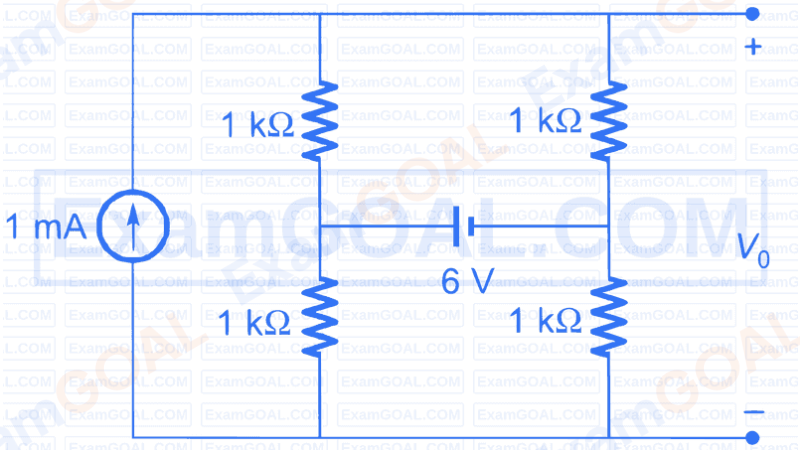
1. **Solve**
   Print it, or work through it on screen.
2. **Check**
   The answer key is on the last pages.
3. **Understand**
   Scan the QR code beside any question for its full solution.

Marks 1 25 questions

Q1. GATE ECE 2026

The output voltage V_o (in Volt) for the network given in the Figure is ____ . (rounded off to two decimal places)

Open solution

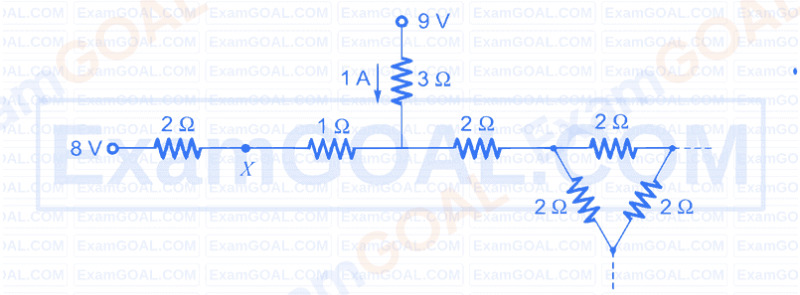


Q2. GATE ECE 2025

Consider a part of an electrical network as shown below. Some node voltages, and the current flowing through the 3Ω resistor are as indicated.

The voltage (in Volts) at node X is _____ .

Open solution

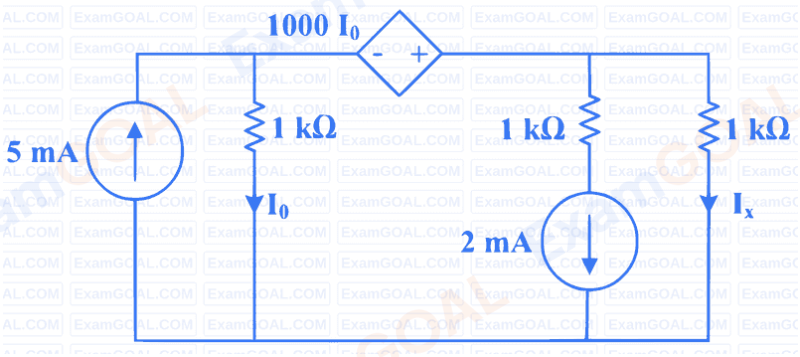


(A) $\frac{20}{3}$ (B) $\frac{32}{3}$ (C) $\frac{22}{3}$ (D) $\frac{2}{3}$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q3. GATE ECE 2024

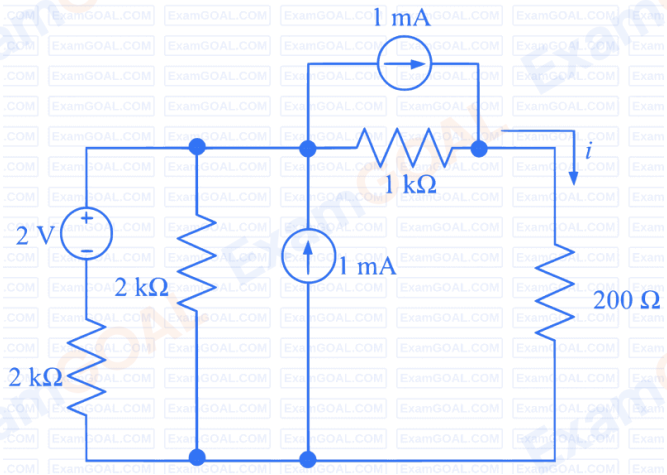
In the given circuit, the current I_x (in mA) is _____ .



Open solution

Q4. GATE ECE 2023

In the circuit shown below, the current i flowing through 200 Ω resistor is _____ mA (rounded off to two decimal places).



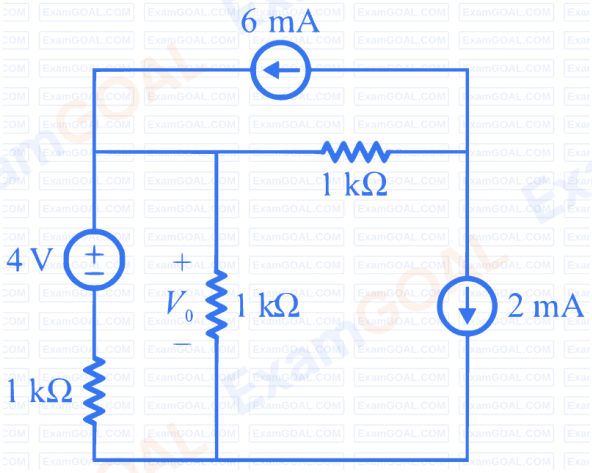
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q5. GATE ECE 2021

Open solution

Consider the circuit shown in the figure.

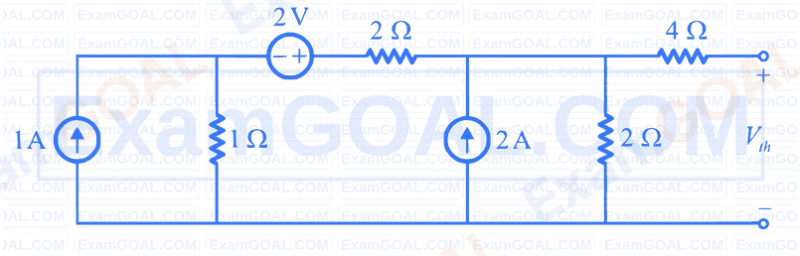


The value of v_0 (rounded off to one decimal place) is _____ V.

Q6. GATE ECE 2020

Open solution

In the circuit shown below, the Thevenin voltage V_{th} is



(A) 4.5 V (B) 3.6 V (C) 2.4 V (D) 2.8 V

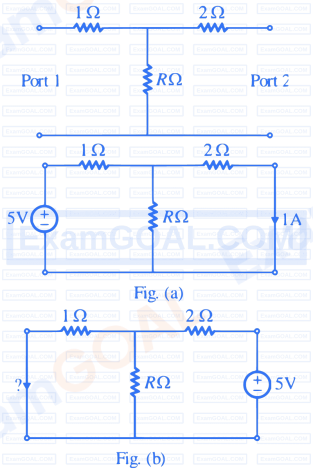
Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q7. GATE ECE 2019

Open solution

Consider the two-port resistive network shown in the figure. When an excitation of 5 V is applied across Port 1, and Port 2 is shorted, the current through the short circuit at Port 2 is measured to be 1 A (see (a) in the figure).

Now, if an excitation of 5 V is applied across Port 2, and Port 1 is shorted (see (b) in the figure), what is the current through the short circuit at Port 1?

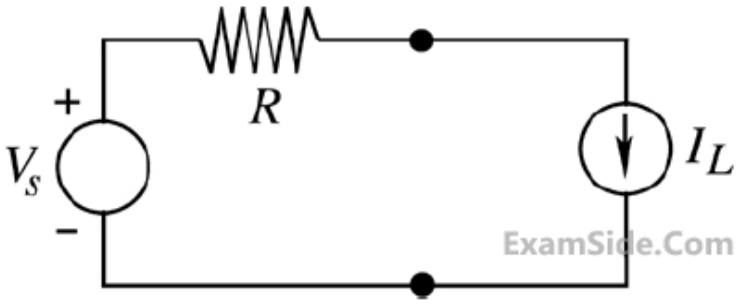


(A) 0.5 A (B) 2.5 A (C) 1 A (D) 2 A

Q8. GATE ECE 2016 Set 2

Open solution

In the circuit shown below, V_S is a constant voltage source and I_L is a constant current load



The value of I_L that maximizes the power absorbed by the constant current load is

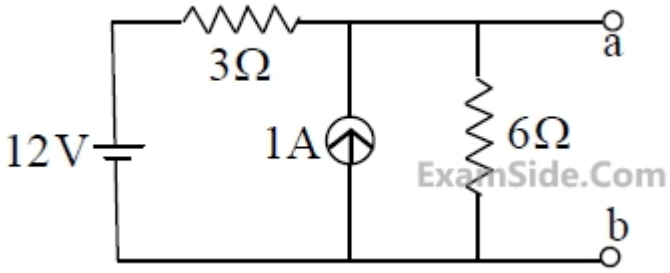
(A) $\frac{V_s}{4R}$ (B) $\frac{V_s}{2R}$ (C) $\frac{V_s}{R}$ (D) ∞

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q9. GATE ECE 2015 Set 3

For the circuit shown in the figure, the Thevenin equivalent voltage (in Volts) across terminals a-b is _____.

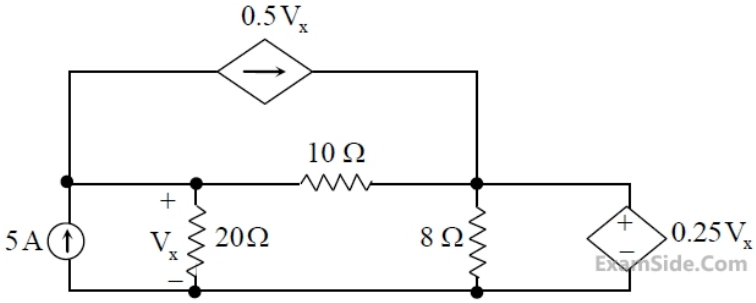
Open solution



Q10. GATE ECE 2015 Set 3

In the circuit shown, the voltage V_x (in Volts) is _____.

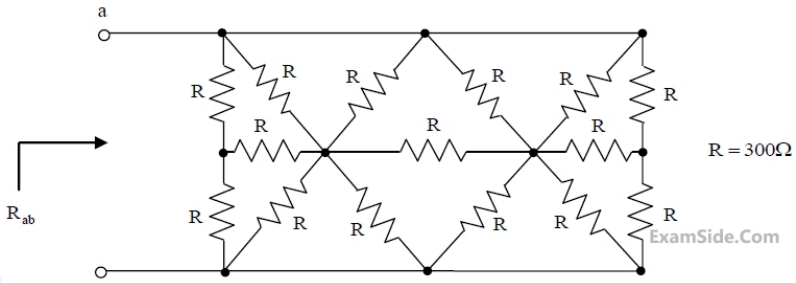
Open solution



Q11. GATE ECE 2015 Set 1

In the network shown in the figure, all resistors are identical with R = 300 Ω. The resistance R_{ab} (in Ω) of the network is _____.

Open solution



Q12. GATE ECE 2014 Set 1

For maximum power transfer between two cascaded sections of an electrical network, the relationship between the output impedance Z_1 of the first section to the input impedance Z_2 of the second section is

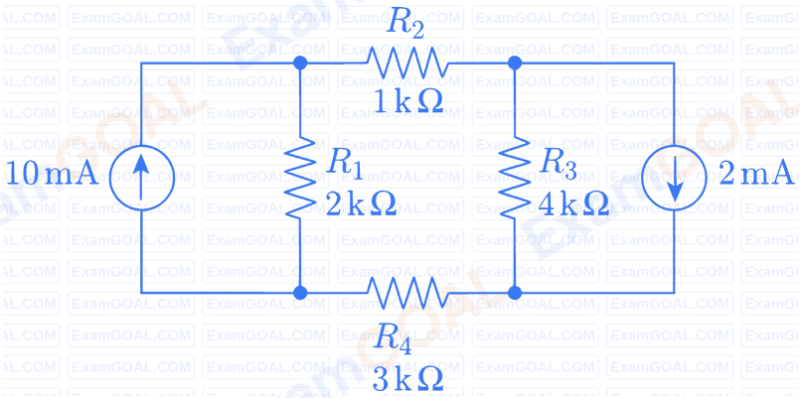
Open solution

(A) $Z_2 = Z_1$ (B) $Z_2 = -Z_1$ (C) $Z_2 = Z_1^*$ (D) $Z_2 = -Z_1^*$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q13. GATE ECE 2014 Set 4

The magnitude of current (in mA) through the resistor R_2 in the figure shown is ____.



Open solution

Q14. GATE ECE 2014 Set 2

Norton's theorem states that a complex network connected to a load can be replaced with an equivalent impedance

(A) in series with a current source
(B) in parallel with a voltage source
(C) in series with a voltage source
(D) in parallel with a current source

Open solution

Q15. GATE ECE 2013

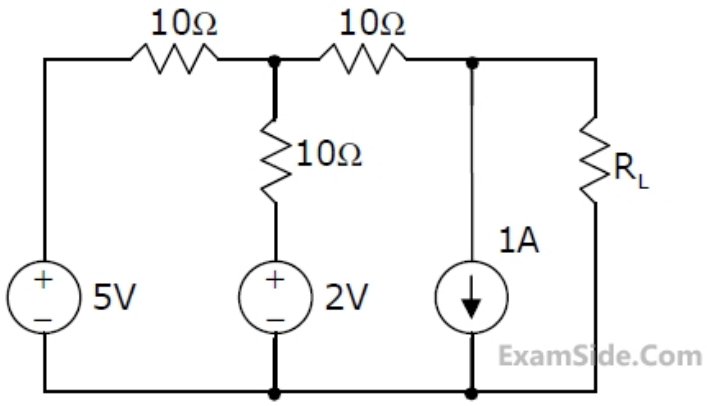
A source $v_s(t) = V \cos 100 \pi t$ has an internal impedance of (4 + j3) Ω. If a purely resistive load connected to this source has to extract the maximum power out of the source, its value in Ω shoud be

(A) 3 (B) 4 (C) 5 (D) 7

Open solution

Q16. GATE ECE 2011

In the circuit shown below the value of R_L such that the power transferred to R_L is maximum



(A) 5Ω (B) 10Ω (C) 15Ω (D) 20Ω

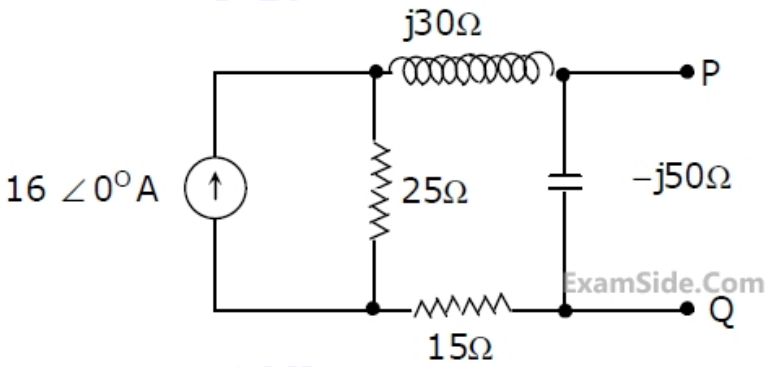
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q17. GATE ECE 2011

In the circuit shown below, the Norton equivalent current in amperes with respect to terminals P and Q is

Open solution

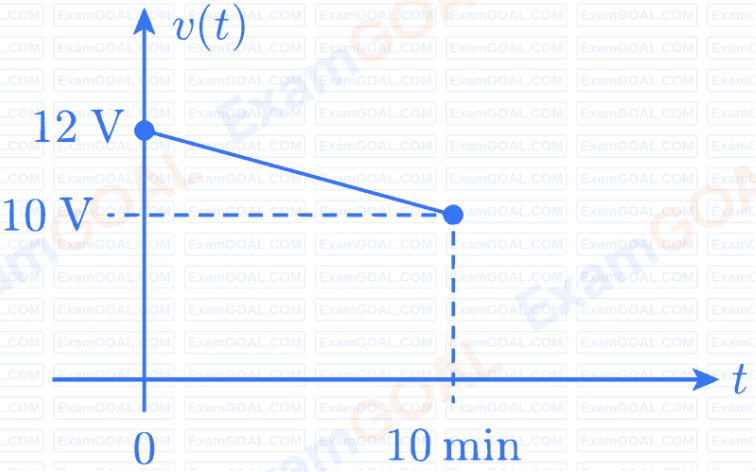


(A) 6.4 - j 4.8 (B) 6.56 - j 7.87 (C) 10 + j 0 (D) 16 + j 0

Q18. GATE ECE 2009

A fully charged mobile phone with a 12 V battery is good for a 10 minute talk-time. Assume that, during the talk-time, the battery delivers a constant current of 2 A and its voltage drops linearly from 12 V to 10 V as shown in the figure. How much energy does the battery deliver during this talk-time?

Open solution



(A) 220 J (B) 12 kJ (C) 13.2 kJ (D) 14.4 kJ

Q19. GATE ECE 2007

An independent voltage source in series with an impedance $Z_s=R_s+jX_s$ delivers a maximum average power to a load impedance Z_L when

Open solution

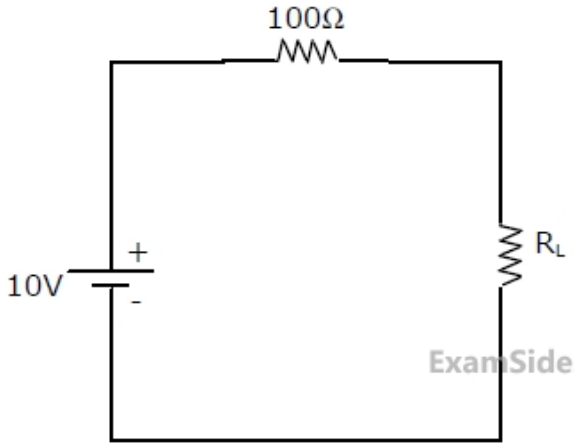
(A) $Z_L=R_s+jX_s$ (B) $Z_L=R_s$ (C) $Z_L=jX_s$ (D) $Z_L=R_s-jX_s$

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q20. GATE ECE 2005

The maximum power that can be transferred to the load resistor R_L from the voltage source in figure is

Open solution



(A) 1 W (B) 10 W (C) 0.25 W (D) 0.5 W

Q21. GATE ECE 2003

A source of angular frequency 1rad/sec has source impedance consisting of 1Ω resistance in series with 1H inductance. The load that will obtain maximum power transfer is

Open solution

(A) 1Ω resistance

(B) 1Ω resistance in parallel with 1H inductance

(C) 1Ω resistance in series with 1F capacitor

(D) 1Ω resistance in parallel with 1F capacitor

Q22. GATE ECE 1998

Superposition theorem is NOT applicable to networks containing

Open solution

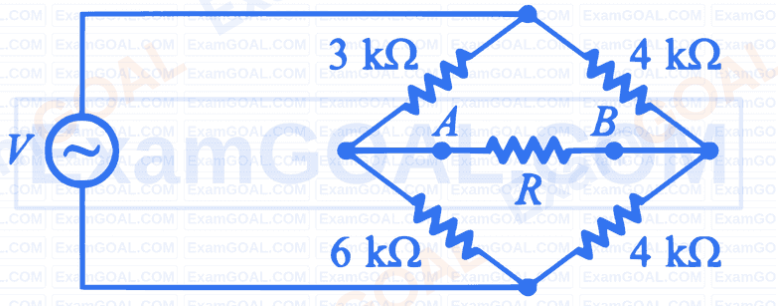
(A) Nonlinear elements (B) Dependent voltage sources

(C) Dependent current sources (D) Transformers

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q23. GATE ECE 1995

The value of the resistance, R, connected across the terminals, A and B, (ref. Fig.), which will absorb the maximum power, is



Open solution

(A) 4.00 kΩ (B) 4.11 kΩ (C) 8.00 kΩ (D) 9.00 kΩ

Q24. GATE ECE 1994

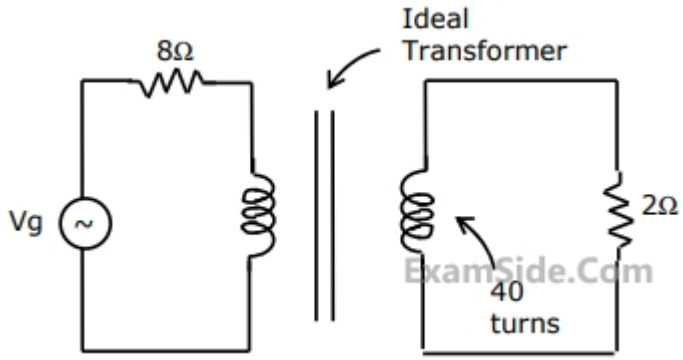
A generator of internal impedance, Z_G, delivers maximum power to a load impedance, Z_L, only if Z_L = ...

Open solution

(A) Z_G^* (B) Z_G (C) $-Z_G$ (D) $2Z_G$

Q25. GATE ECE 1993

If the secondary winding of the ideal transformer shown in the circuit of the figure has 40 turns, the number of turns in the primary winding for maximum power transfer to the 2 Ω resistor will be



Open solution

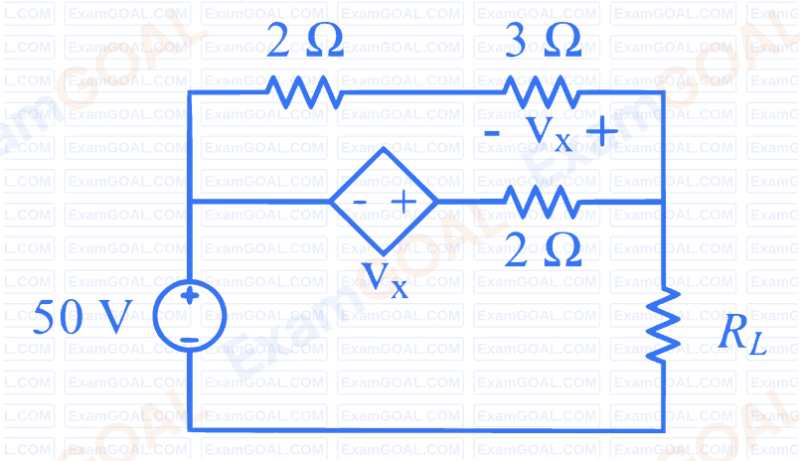
(A) 20 (B) 40 (C) 80 (D) 160

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Marks 2 26 questions

Q26. GATE ECE 2024

In the network shown below, maximum power is to be transferred to the load R_L.

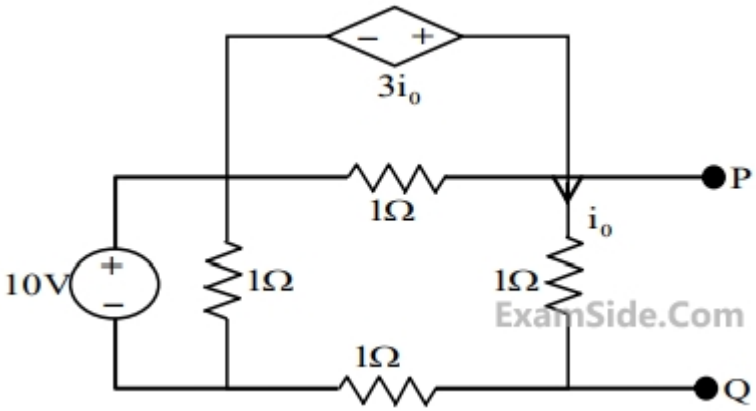


The value of R_L (in Ω) is _____.

Open solution

Q27. GATE ECE 2017 Set 2

Consider the circuit shown in the figure.

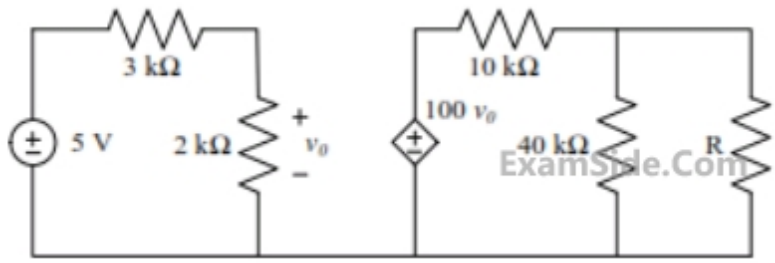


The Thevenin equivalent resistance (in Ω) across P – Q is _____.

Open solution

Q28. GATE ECE 2016 Set 1

In the circuit shown in the figure, the maximum power (in watt) delivered to the resistor R is _____.



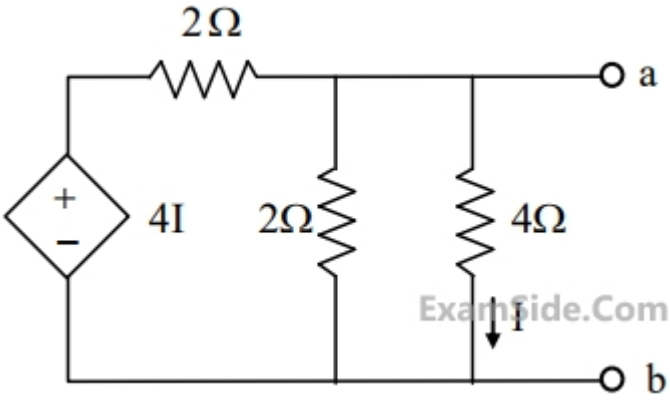
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q29. GATE ECE 2015 Set 2

Open solution

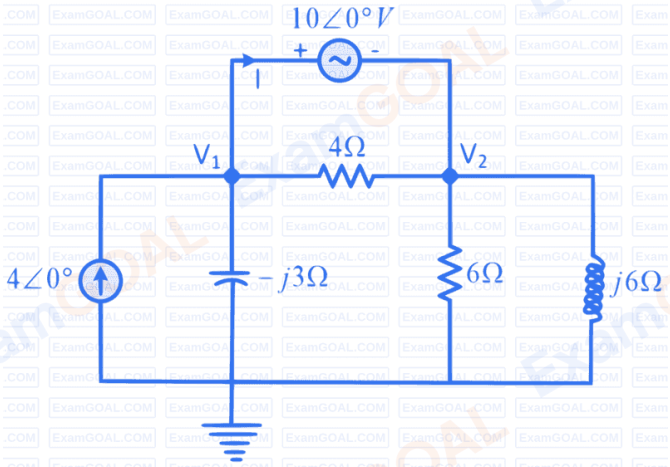
In the circuit shown, the Norton equivalent resistance (in Ω) across terminals a–b is _____.



Q30. GATE ECE 2014 Set 3

Open solution

In the circuit shown in the figure, the value of node voltage V_2 is

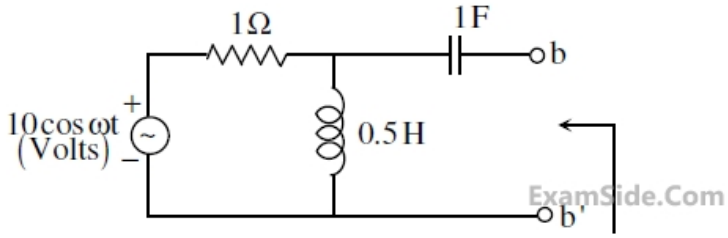


(A) 22 + j 2 V (B) 2 + j 22 V (C) 22 - j 2 V (D) 2 - j 22 V

Q31. GATE ECE 2014 Set 3

Open solution

In the circuit shown in the figure, the angular frequency ω (in rad/s), at which the Norton equivalent impedance as seen from terminals b-b' is purely resistive, is _____.

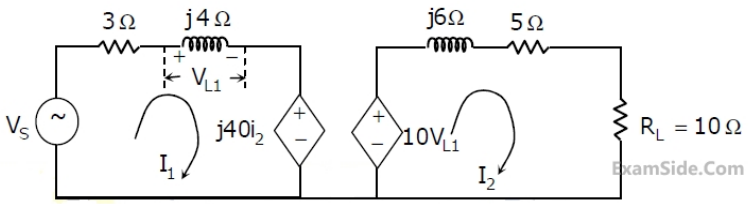


Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q32. GATE ECE 2013

Open solution

In the circuit shown below, if the source voltage $V_s = 100\angle 53.13^0 V$ then the Thevenin's equivalent voltage in Volts as seen by the load resistance R_L is

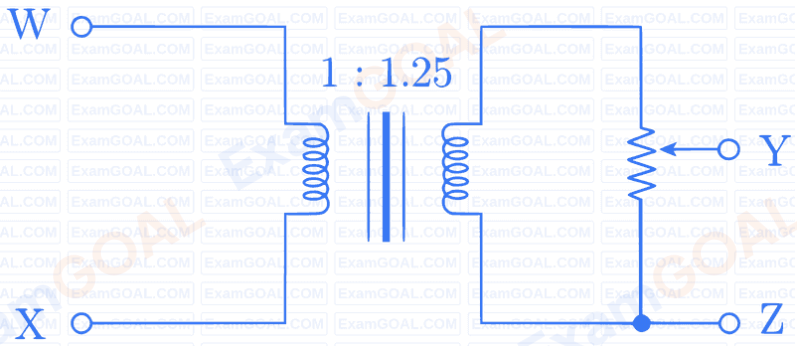


(A) $100\angle 90^0$ (B) $800\angle 0^0$ (C) $800\angle 90^0$ (D) $100\angle 60^0$

Q33. GATE ECE 2013

Open solution

The following arrangement consists of an ideal transformer and an attenuator which attenuates by a factor of 0.8. An ac voltage V_{wx1} = 100V is applied across WX to get an open circuit voltage YZ1 V across YZ. Next, an ac voltage V_{YZ2} =100V is applied across YZ to get an open circuit voltage V_{WX2} across WX. Then, V_{YZ1} / V_{WX1} , V_{WX2} / V_{YZ2} are respectively,



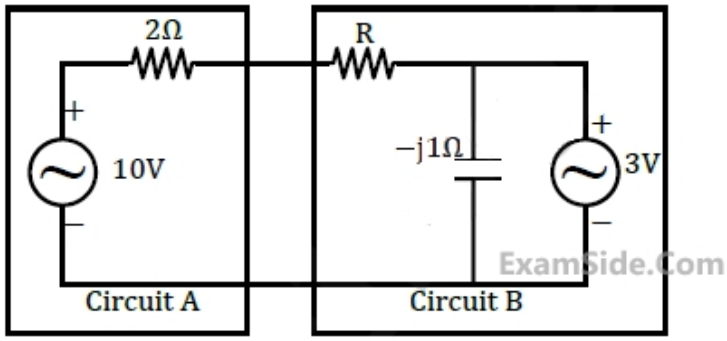
(A) 125 / 100 and 80 / 100
(B) 100 / 100 and 80 / 100
(C) 100 / 100 and 100 / 100
(D) 80 / 100 and 80 / 100

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q34. GATE ECE 2012

Open solution

Assuming both the voltage sources are in phase the value of R for which maximum power is transferred from circuit A to circuit B is

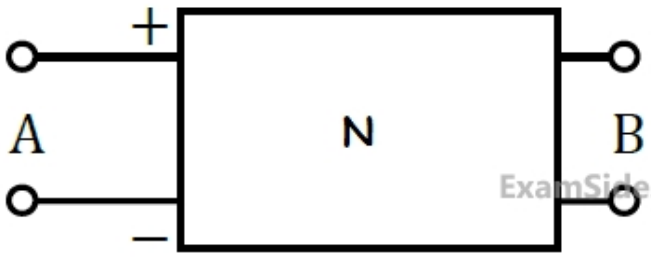


(A) 0.8Ω (B) 1.4Ω (C) 2Ω (D) 2.8Ω

Q35. GATE ECE 2012

Open solution

With 10 V dc connected at port A in the linear nonreciprocal two-port network shown below, the following were observed:
(i) 1Ω connected at port B draws a current of 3 A
(ii) 2.5Ω connected at port B draws a current of 2 A



For the same network, with 6 V dc connected at port A, 1Ω connected at port B draws 7/3 A. If 8 V dc is connected to port A, the open circuit voltage at port B is

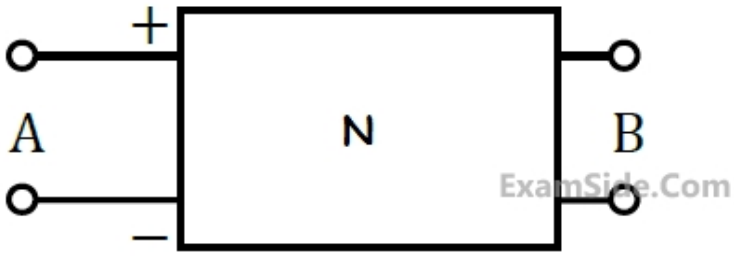
(A) 6 V (B) 7 V (C) 8 V (D) 9 V

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q36. GATE ECE 2012

Open solution

With 10 V dc connected at port A in the linear nonreciprocal two-port network shown below, the following were observed:
(i) 1Ω connected at port B draws a current of 3 A
(ii) 2.5Ω connected at port B draws a current of 2 A



ExamSide.Com

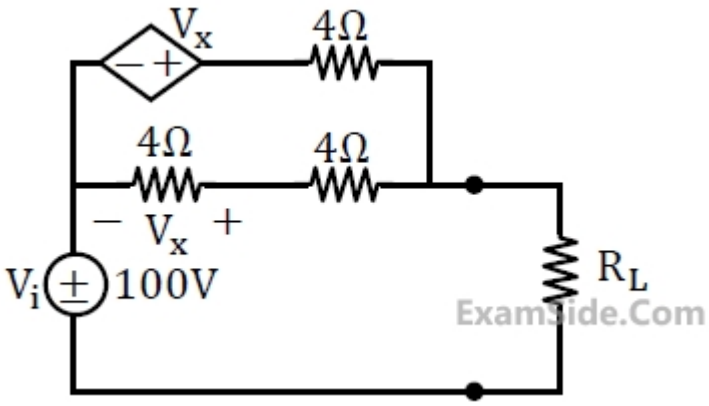
With 10 V dc connected at port A, the current drawn by 7Ω connected at port B is

(A) 3/7 A (B) 5/7 A (C) 1 A (D) 9/7 A

Q37. GATE ECE 2009

Open solution

In the circuit shown, what value of R_L maximizes the power delivered to R_L?



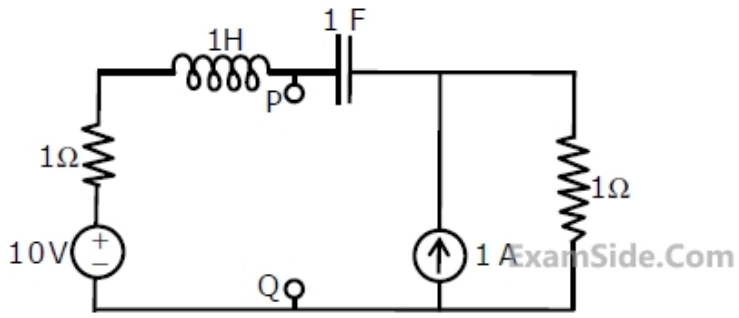
ExamSide.Com

(A) 2.4Ω (B) $\frac{8}{3}\Omega$ (C) 4Ω (D) 6Ω

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q38. GATE ECE 2008

The Thevenin equivalent impedance Z_{th} between the nodes P and Q in the following circuit is

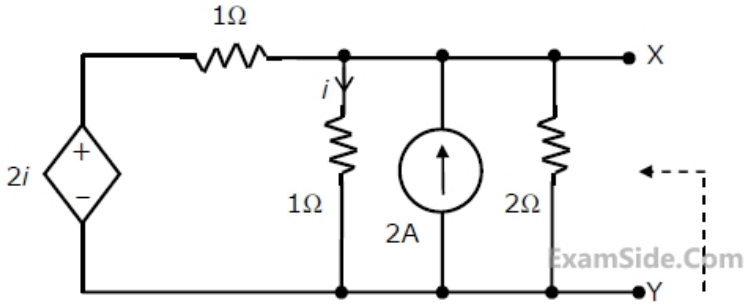


Open solution

(A) 1 (B) $1+s+\frac{1}{s}$ (C) $2+s+\frac{1}{s}$ (D) $\frac{s^2+s+1}{s^2+2s+1}$

Q39. GATE ECE 2007

For the circuit shown in the figure, the Thevenin voltage and resistance looking into X-Y are:

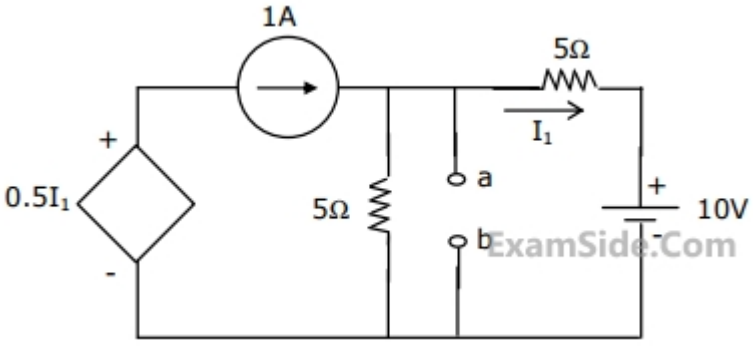


Open solution

(A) $\frac{4}{3}V, 2\Omega$ (B) $4V, \frac{2}{3}\Omega$ (C) $\frac{4}{3}V, \frac{2}{3}\Omega$ (D) $4V, 2\Omega$

Q40. GATE ECE 2005

For the circuit shown in figure, Thevenin's voltage and Thevenin's equivalent resistance at terminals a – b is



Open solution

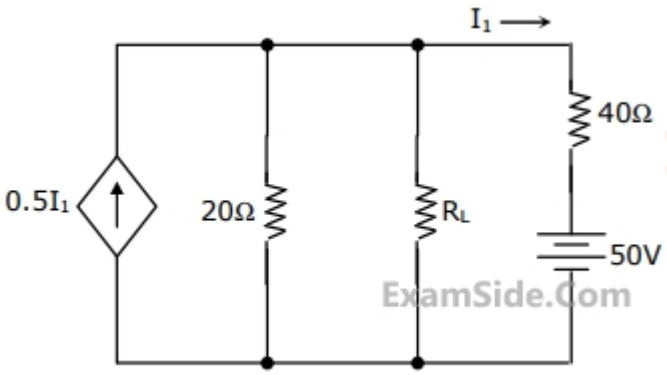
(A) 5 V and 2 Ω (B) 7.5 V and 2.5 Ω (C) 4 V and 2 Ω (D) 3 V and 2.5 Ω

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q41. GATE ECE 2002

Open solution

In the network of Figure, the maximum power is delivered to R_L if its value is

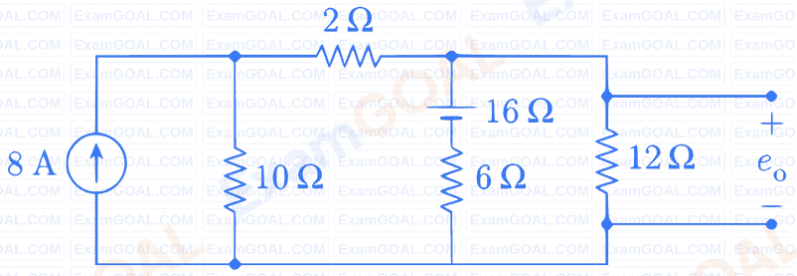


(A) 16 Ω (B) (40/3) Ω (C) 60 Ω (D) 20 Ω

Q42. GATE ECE 2001

Open solution

The voltage e_0 in figure is

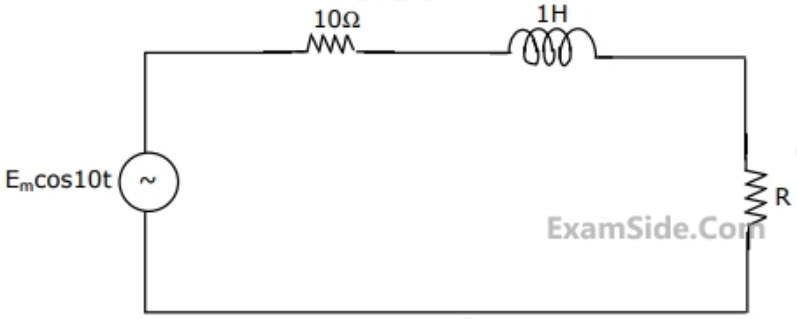


(A) 48 V (B) 24 V (C) 36 V (D) 28 V

Q43. GATE ECE 2001

Open solution

In figure, the value of the load resistor R which maximizes the power delivered to it is

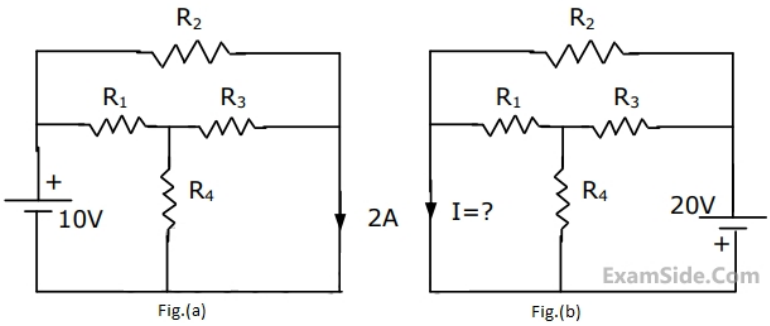


(A) 14.14 Ω (B) 10 Ω (C) 200 Ω (D) 28.28 Ω

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q44. GATE ECE 2000

Use the data of Fig.(a). The current i in the circuit of Fig.(b) is

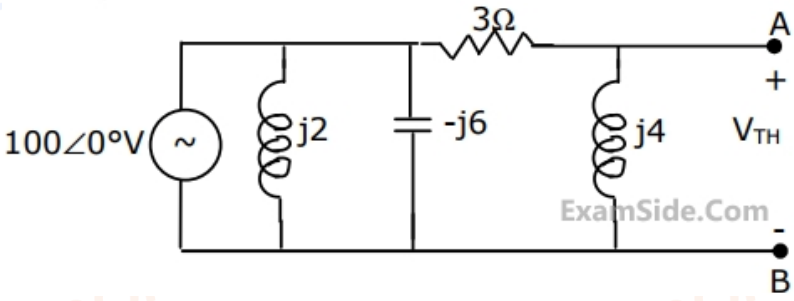


Open solution

(A) -2 A (B) 2 A (C) -4 A (D) +4 A

Q45. GATE ECE 1999

The Thevenin equivalent voltage V_{TH} appearing between the terminals A and B of the network shown in Fig. is given by

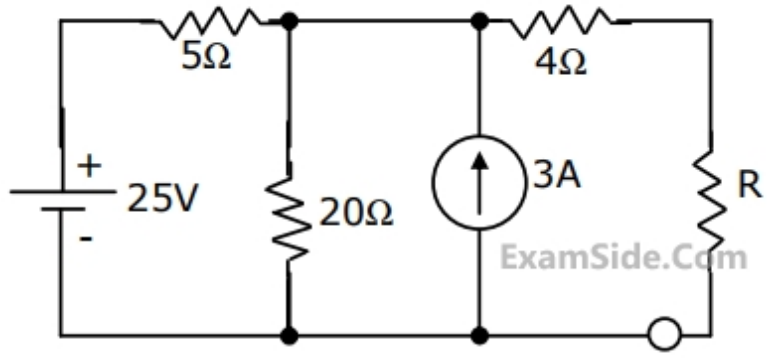


Open solution

(A) j16(3 - j4) V (B) j16(3 + j4) V (C) 16(3 + j4) V (D) 16(3 - j4) V

Q46. GATE ECE 1999

The value of R (in ohms) required for maximum power transfer in the network shown in Fig. is



Open solution

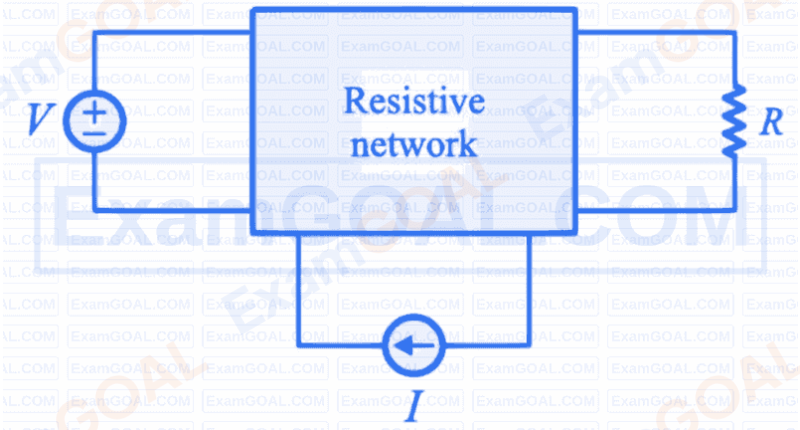
(A) 2 (B) 4 (C) 8 (D) 16

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q47. GATE ECE 1995

Open solution

A DC circuit shown in the figure has a voltage source **V**, a current source **I** and several resistors. A particular resistor **R** dissipates a power of 4 Watts when **V** alone is active. The same resistor **R** dissipates a power of 9 Watts when **I** alone is active. The power dissipated by **R** when both sources are active will be

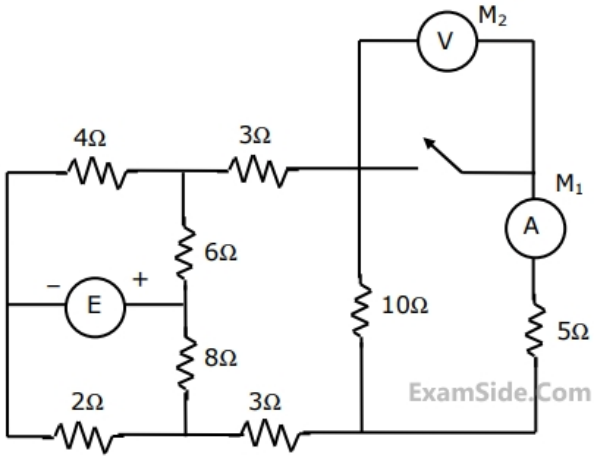


(A) 1 W (B) 5 W (C) 13 W (D) 25 W

Q48. GATE ECE 1993

Open solution

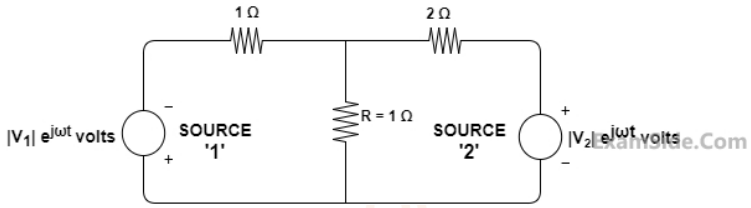
In the circuit of figure, when switch S1 is closed, the ideal ammeter M_1 reads 5A. What will be ideal voltmeter M_2 read when S_1 is kept open? (The value of E is not specified).



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q49. GATE ECE 1989

In the circuit of figure, the power dissipated in the register R is 1W when only source '1' is present and '2' replaced by a short.The power dissipated in the same registor R is 4W when only source '2' is present and '1' is replaced by a short.When both the source '1' and '2' are present, the power dissipated in R will be:



(A) 1 W (B) 3 W (C) 4 W (D) 5 W

Open solution

Q50. GATE ECE 1989

A load, $Z_L = R_L + jX_L$ is to be matched, using an ideal transformer, to a generator of internal impedance, $Z_S = R_S + jX_S$. The turns ratio of the transformer required is

(A) $\sqrt{|Z_S \,/\, Z_L|}$ (B) $\sqrt{|R_L \,/\, R_S|}$

(C) $\sqrt{|R_L \,/\, Z_S|}$ (D) $\sqrt{|Z_L \,/\, R_S|}$

Open solution

Q51. GATE ECE 1988

If an impedance Z_L is connected across a voltage source V with source impedance Z_S, then for maximum power transfer, the load impedance must be equal to

(A) Source impedance Z_S (B) Complex conjugate of Z_S

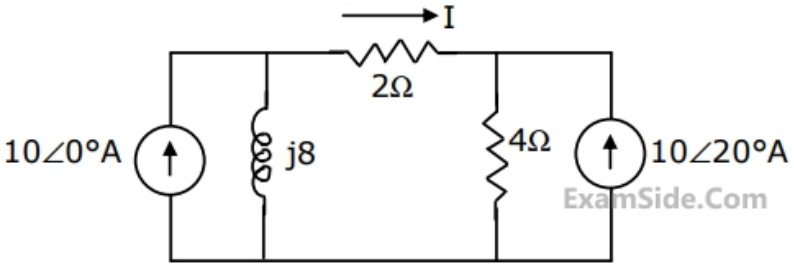
(C) Real part of Z_S (D) Imaginary part of Z_S

Open solution

Marks 5 5 questions

Q52. GATE ECE 1999

For the network shown in Fig., evaluate the current I flowing through the 2Ω resistor using superposition theorem.



Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

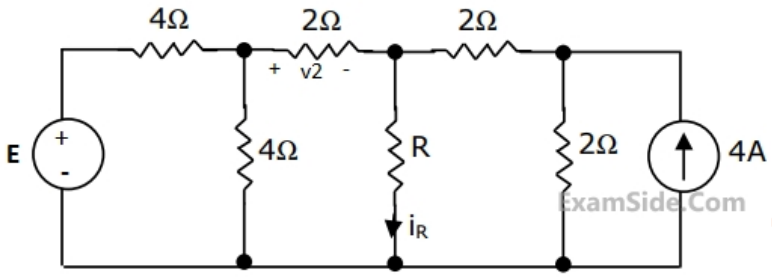
Q53. GATE ECE 1998

A voltage source of internal impedance $R_s + jX_s$ supplies power to a load of impedance $R_L + jX_L$ in which only R_L is variable. Determine the value of R_L for maximum power transfer from the source to the load. Also, find the numerical value of R_L if the source impedance is 3.0 Ω (purely resistive) and X_L is 4.0 Ω.

Open solution

Q54. GATE ECE 1997

In the circuit of Fig. when R = 0 Ω, the current i_R equals 10 A.



(a) Find the value of R for which it absorbs maximum power.

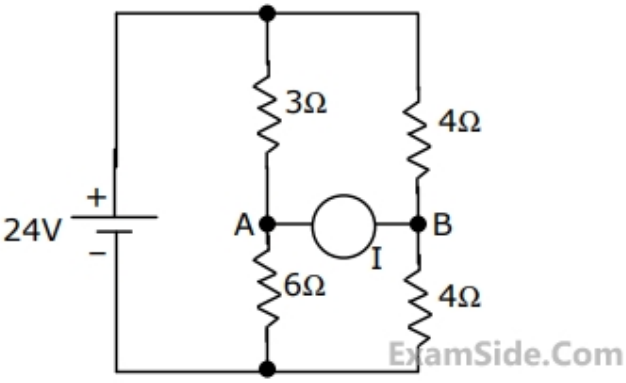
(b) Find the value of E.

(c) Find v2 when R = ∞ (open circuit)

Open solution

Q55. GATE ECE 1996

In the circuit shown in Fig., it is known that the variable current source I absorbs power.Find I (in magnitude and direction) so that it receives maximum power and also find the amount of power absorbed by it.



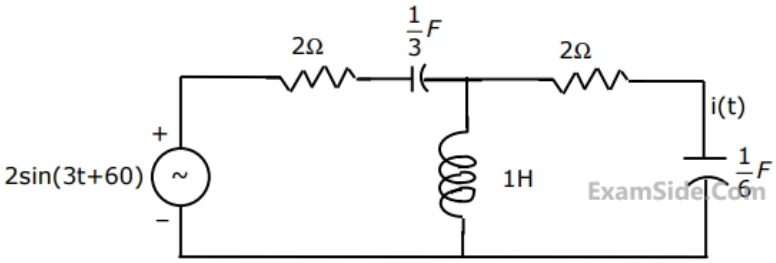
Open solution

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Q56. GATE ECE 1995

Open solution

Determine the current, i(t), in the circuit given below, (Fig.), using the Thevenin's theorem.



Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

Answer Key

Marks 1

1. **(1) (0.95 to 1.05)**	2. **A**	3. **2**	4. **1.30 to 1.40**	5. **1**
6. **B**	7. **C**	8. **B**	9. **10**	10. **8**
11. **100**	12. **C**	13. **2.8**	14. **D**	15. **C**
16. **C**	17. **A**	18. **C**	19. **D**	20. **C**
21. **C**	22. **A**	23. **A**	24. **A**	25. **C**

Marks 2

26. **2.5**	27. **-1**	28. **0.8**	29. **1.33**	30. **D**
31. **2**	32. **C**	33. **B**	34. **A**	35. **C**
36. **C**	37. **C**	38. **A**	39. **D**	40. **B**
41. **A**	42. **D**	43. **A**	44. **C**	45. **A**
46. **C**	47. **A, D**	48. **50**	49. **A**	50. **A**
51. **B**				

Marks 5

52. $\mathrm{I} = 7.61\angle -113.8^\circ$	53. $\mathrm{R_L} = \sqrt{\mathrm{R_S^2} + (\mathrm{X_L} + \mathrm{X_S})^2}$ $\mathrm{R_L}$ **=5 Ω**	54. **(a) R = 2 Ω** **(b) E = 64 V** **(c) v2 = 6 V**	55. **I = 0.5 A from A to B, P_{max} = 1 W**	56. **i(t) = 0.96** $\sin(3t + 31.4^\circ)$

For numerical answers given as a range, any value within the range is correct. Full solutions: scan the QR code next to each question, or visit questions.examside.com.

Keep practising on the ExamGOAL app
Chapter-wise PYQs with solutions, full mock tests and analysis for JEE, NEET, GATE and more.
GET IT ON Google Play

ExamGOAL
PYQ PDF

Your Exam. Our Responsibility.

From first chapter to exam day.

Practise with the questions your exam has already asked, then test yourself under real exam conditions.

PAST YEAR QUESTIONS

Every past paper, chapter-wise.

Previous year questions sorted the way you study, each with a full solution.

- Chapter-wise and year-wise
- Weightage and topic tags on every chapter
- Your progress on each one, at a glance

MOCK TESTS

Mock tests with a real exam feel.

Practise under the same pressure as the real paper, on your phone or on the big screen.

- Timed sections, just like exam day
- Mark questions for review and come back

ANALYSIS

Know your score. Know your rank.

A full report after every test, so you know exactly where you stand.

- Your percentile among everyone who took it
- Predicted real-exam percentile and All India Rank
- Subject breakdown and time analysis

PRACTICE

Solve. Check. Understand.

Learn from every question, not just the ones you get right.

- Instant answers as you practise
- Step-by-step explanations
- See how everyone else answered

NCERT Exemplar
With full solutions

Exam syllabus
Subject and chapter-wise

Bookmarks
Organised in groups

Your notes
On any question

ExamGOAL
LEARNING APP

ExamGOAL: Exam Preparation App

One app for every exam. Free trial for new users, on Android and the web.

GET IT ON Google Play

Scan to get the app

www.examgoal.com
Practice and mock tests

questions.examside.com
Free solutions

pdf.questions.examgoal.com
Free PYQ PDFs